

2. The problem is to solve the differential equation

$$a_n \frac{d^n}{dt^n} u(t) + a_{n-1} \frac{d^{n-1}}{dt^{n-1}} u(t) + \cdots + a_0 u(t) = f(t),$$

where $a_0, a_1, \dots, a_n$ are complex constants, and f is a given function. Here we suppose that f has bounded support and is smooth (say of class C^2).

(a) Let

$$\hat{f}(z) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i z t} dt.$$

Observe that $\hat{f}$ is an entire function, and using integration by parts show that

$$|\hat{f}(x + iy)| \leq \frac{A}{1 + x^2}$$

if $|y| \leq a$ for any fixed $a \geq 0$.

(b) Write

$$P(z) = a_n (2\pi i z)^n + a_{n-1} (2\pi i z)^{n-1} + \cdots + a_0.$$

Find a real number c so that $P(z)$ does not vanish on the line

$$L = \{z : z = x + ic, \ x \in \mathbb{R}\}.$$

(c) Set

$$u(t) = \int_L \frac{e^{2\pi i z t}}{P(z)} \hat{f}(z) dz.$$

Check that

$$\sum_{j=0}^n a_j \left(\frac{d}{dt} \right)^j u(t) = \int_L e^{2\pi i z t} \hat{f}(z) dz$$

and

$$\int_L e^{2\pi i z t} \hat{f}(z) dz = \int_{-\infty}^{\infty} e^{2\pi i x t} \hat{f}(x) dx.$$

Conclude by the Fourier inversion theorem that

$$\sum_{j=0}^n a_j \left(\frac{d}{dt} \right)^j u(t) = f(t).$$

Note that the solution u depends on the choice c .

3.* In this problem, we investigate the behavior of certain bounded holomorphic functions in an infinite strip. The particular result described here is sometimes called the three-lines lemma.

- (a) Suppose $F(z)$ is holomorphic and bounded in the strip $0 < \operatorname{Im}(z) < 1$ and continuous on its closure. If $|F(z)| \leq 1$ on the boundary lines, then $|F(z)| \leq 1$ throughout the strip.
- (b) For the more general F , let $\sup_{x \in \mathbb{R}} |F(x)| = M_0$ and $\sup_{x \in \mathbb{R}} |F(x+i)| = M_1$. Then,

$$\sup_{x \in \mathbb{R}} |F(x+iy)| \leq M_0^{1-y} M_1^y, \quad \text{if } 0 \leq y \leq 1.$$

- (c) As a consequence, prove that $\log \sup_{x \in \mathbb{R}} |F(x+iy)|$ is a convex function of y when $0 \leq y \leq 1$.

[Hint: For part (a), apply the maximum modulus principle to $F_\epsilon(z) = F(z)e^{-\epsilon z^2}$. For part (b), consider $M_0^{z-1} M_1^{-z} F(z)$.]

4.* There is a relation between the Paley-Wiener theorem and an earlier representation due to E. Borel.

- (a) A function $f(z)$, holomorphic for all z , satisfies $|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}$ for all ϵ if and only if it is representable in the form

$$f(z) = \int_C e^{2\pi i z w} g(w) dw$$

where g is holomorphic outside the circle of radius M centered at the origin, and g vanishes at infinity. Here C is any circle centered at the origin of radius larger than M . In fact, if $f(z) = \sum a_n z^n$, then $g(w) = \sum_{n=0}^{\infty} A_n w^{-n-1}$ with $a_n = A_n (2\pi i)^{n+1} / n!$.

- (b) The connection with Theorem 3.3 is as follows. For these functions f (for which in addition f and $\hat{f}$ are of moderate decrease on the real axis), one can assert that the g above is holomorphic in the larger region, which consists of the slit plane $\mathbb{C} - [-M, M]$. Moreover, the relation between g and the Fourier transform $\hat{f}$ is

$$g(z) = \frac{1}{2\pi i} \int_{-M}^M \frac{\hat{f}(\xi)}{\xi - z} d\xi$$

so that $\hat{f}$ represents the jump of g across the segment $[-M, M]$; that is,

$$\hat{f}(x) = \lim_{\epsilon \rightarrow 0, \epsilon > 0} g(x+i\epsilon) - g(x-i\epsilon).$$

See Problem 5 in Chapter 3.

5 Entire Functions

...but after the 15th of October I felt myself a free man, with such longing for mathematical work, that the last two months flew by quickly, and that only today I found the letter of the 19th of October that I had not answered. The result of my work, with which I am not entirely satisfied, I want to share with you.

Firstly, in looking back at my lectures, a gap in function theory needed to be filled. As you know, up to now the following question had been unresolved. Given an arbitrary sequence of complex numbers, $a_1, a_2, \dots$, can one construct an entire (transcendental) function that vanishes at these values, with prescribed multiplicities, and nowhere else?...

K. Weierstrass, 1874

In this chapter, we will study functions that are holomorphic in the whole complex plane; these are called **entire** functions. Our presentation will be organized around the following three questions:

1. Where can such functions vanish? We shall see that the obvious necessary condition is also sufficient: if $\{z_n\}$ is any sequence of complex numbers having no limit point in $\mathbb{C}$, then there exists an entire function vanishing exactly at the points of this sequence. The construction of the desired function is inspired by Euler's product formula for $\sin \pi z$ (the prototypical case when $\{z_n\}$ is $\mathbb{Z}$), but requires an additional refinement: the Weierstrass canonical factors.
2. How do these functions grow at infinity? Here, matters are controlled by an important principle: the larger a function is, the more zeros it can have. This principle already manifests itself in the simple case of polynomials. By the fundamental theorem of algebra, the number of zeros of a polynomial P of degree d is precisely d , which is also the exponent in the order of (polynomial) growth of P , namely

$$\sup_{|z|=R} |P(z)| \approx R^d \quad \text{as } R \rightarrow \infty.$$

A precise version of this general principle is contained in Jensen's formula, which we prove in the first section. This formula, central to much of the theory developed in this chapter, exhibits a deep connection between the number of zeros of a function in a disc and the (logarithmic) average of the function over the circle. In fact, Jensen's formula not only constitutes a natural starting point for us, but also leads to the fruitful theory of value distributions, also called Nevanlinna theory (which, however, we do not take up here).

3. To what extent are these functions determined by their zeros? It turns out that if an entire function has a finite (exponential) order of growth, then it can be specified by its zeros up to multiplication by a simple factor. The precise version of this assertion is the Hadamard factorization theorem. It may be viewed as another instance of the general rule that was formulated in Chapter 3, that is, that under appropriate conditions, a holomorphic function is essentially determined by its zeros.

1 Jensen's formula

In this section, we denote by D_R and C_R the open disc and circle of radius R centered at the origin. We shall also, in the rest of this chapter, exclude the trivial case of the function that vanishes identically.

Theorem 1.1 *Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω , $f(0) \neq 0$, and f vanishes nowhere on the circle C_R . If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities),¹ then*

$$(1) \quad \log |f(0)| = \sum_{k=1}^N \log \left(\frac{|z_k|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta.$$

The proof of the theorem consists of several steps.

Step 1. First, we observe that if f_1 and f_2 are two functions satisfying the hypotheses and the conclusion of the theorem, then the product $f_1 f_2$ also satisfies the hypothesis of the theorem and formula (1). This observation is a simple consequence of the fact that $\log xy = \log x + \log y$ whenever x and y are positive numbers, and that the set of zeros of $f_1 f_2$ is the union of the sets of zeros of f_1 and f_2 .

¹That is, each zero appears in the sequence as many times as its order.

Step 2. The function

$$g(z) = \frac{f(z)}{(z - z_1) \cdots (z - z_N)}$$

initially defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j . Therefore each z_j is a removable singularity, and hence we can write

$$f(z) = (z - z_1) \cdots (z - z_N)g(z)$$

where g is holomorphic in Ω and nowhere vanishing in the closure of D_R . By Step 1, it suffices to prove Jensen's formula for functions like g that vanish nowhere, and for functions of the form $z - z_j$.

Step 3. We first prove (1) for a function g that vanishes nowhere in the closure of D_R . More precisely, we must establish the following identity:

$$\log |g(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| d\theta.$$

In a slightly larger disc, we can write $g(z) = e^{h(z)}$ where h is holomorphic in that disc. This is possible since discs are simply connected, and we can define $h = \log g$ (see Theorem 6.2 in Chapter 3). Now we observe that

$$|g(z)| = |e^{h(z)}| = |e^{\operatorname{Re}(h(z)) + i \operatorname{Im}(h(z))}| = e^{\operatorname{Re}(h(z))},$$

so that $\log |g(z)| = \operatorname{Re}(h(z))$. The mean value property (Corollary 7.3 in Chapter 3) then immediately implies the desired formula for g .

Step 4. The last step is to prove the formula for functions of the form $f(z) = z - w$, where $w \in D_R$. That is, we must show that

$$\log |w| = \log \left(\frac{|w|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |Re^{i\theta} - w| d\theta.$$

Since $\log(|w|/R) = \log |w| - \log R$ and $\log |Re^{i\theta} - w| = \log R + \log |e^{i\theta} - w/R|$, it suffices to prove that

$$\int_0^{2\pi} \log |e^{i\theta} - a| d\theta = 0, \quad \text{whenever } |a| < 1.$$

This in turn is equivalent (after the change of variables $\theta \mapsto -\theta$) to

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0, \quad \text{whenever } |a| < 1.$$

To prove this, we use the function $F(z) = 1 - az$, which vanishes nowhere in the closure of the unit disc. As a consequence, there exists a holomorphic function G in a disc of radius greater than 1 such that $F(z) = e^{G(z)}$. Then $|F| = e^{\operatorname{Re}(G)}$, and therefore $\log |F| = \operatorname{Re}(G)$. Since $F(0) = 1$ we have $\log |F(0)| = 0$, and an application of the mean value property (Corollary 7.3 in Chapter 3) to the harmonic function $\log |F(z)|$ concludes the proof of the theorem.

From Jensen's formula we can derive an identity linking the growth of a holomorphic function with its number of zeros inside a disc. If f is a holomorphic function on the closure of a disc D_R , we denote by $\mathbf{n}(r)$ (or $\mathbf{n}_f(r)$ when it is necessary to keep track of the function in question) the number of zeros of f (counted with their multiplicities) inside the disc D_r , with $0 < r < R$. A simple but useful observation is that $\mathbf{n}(r)$ is a non-decreasing function of r .

We claim that if $f(0) \neq 0$, and f does not vanish on the circle C_R , then

$$(2) \quad \int_0^R \mathbf{n}(r) \frac{dr}{r} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

This formula is immediate from Jensen's equality and the next lemma.

Lemma 1.2 *If $z_1, \dots, z_N$ are the zeros of f inside the disc D_R , then*

$$\int_0^R \mathbf{n}(r) \frac{dr}{r} = \sum_{k=1}^N \log \left| \frac{R}{z_k} \right|.$$

Proof. First we have

$$\sum_{k=1}^N \log \left| \frac{R}{z_k} \right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}.$$

If we define the characteristic function

$$\eta_k(r) = \begin{cases} 1 & \text{if } r > |z_k|, \\ 0 & \text{if } r \leq |z_k|, \end{cases}$$

then $\sum_{k=1}^N \eta_k(r) = \mathbf{n}(r)$, and the lemma is proved using

$$\sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r} = \sum_{k=1}^N \int_0^R \eta_k(r) \frac{dr}{r} = \int_0^R \left(\sum_{k=1}^N \eta_k(r) \right) \frac{dr}{r} = \int_0^R \mathbf{n}(r) \frac{dr}{r}.$$

2 Functions of finite order

Let f be an entire function. If there exist a positive number ρ and constants $A, B > 0$ such that

$$|f(z)| \leq Ae^{B|z|^\rho} \quad \text{for all } z \in \mathbb{C},$$

then we say that f has **an order of growth** $\leq \rho$. We define *the order of growth of f* as

$$\rho_f = \inf \rho,$$

where the infimum is over all $\rho > 0$ such that f has an order of growth $\leq \rho$.

For example, the order of growth of the function e^{z^2} is 2.

Theorem 2.1 *If f is an entire function that has an order of growth $\leq \rho$, then:*

- (i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r .
- (ii) If $z_1, z_2, \dots$ denote the zeros of f , with $z_k \neq 0$, then for all $s > \rho$ we have

$$\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty.$$

Proof. It suffices to prove the estimate for $n(r)$ when $f(0) \neq 0$. Indeed, consider the function $F(z) = f(z)/z^\ell$ where ℓ is the order of the zero of f at the origin. Then $n_f(r)$ and $n_F(r)$ differ only by a constant, and F also has an order of growth $\leq \rho$.

If $f(0) \neq 0$ we may use formula (2), namely

$$\int_0^R n(x) \frac{dx}{x} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

Choosing $R = 2r$, this formula implies

$$\int_r^{2r} n(x) \frac{dx}{x} \leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

On the one hand, since $n(r)$ is increasing, we have

$$\int_r^{2r} n(x) \frac{dx}{x} \geq n(r) \int_r^{2r} \frac{dx}{x} = n(r) [\log 2r - \log r] = n(r) \log 2,$$

and on the other hand, the growth condition on f gives

$$\int_0^{2\pi} \log |f(Re^{i\theta})| d\theta \leq \int_0^{2\pi} \log |Ae^{BR^\rho}| d\theta \leq C' r^\rho$$

for all large r . Consequently, $n(r) \leq Cr^\rho$ for an appropriate $C > 0$ and all sufficiently large r .

The following estimates prove the second part of the theorem:

$$\begin{aligned} \sum_{|z_k| \geq 1} |z_k|^{-s} &= \sum_{j=0}^{\infty} \left(\sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \right) \\ &\leq \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \\ &\leq c \sum_{j=0}^{\infty} 2^{-js} 2^{(j+1)\rho} \\ &\leq c' \sum_{j=0}^{\infty} (2^{\rho-s})^j \\ &< \infty. \end{aligned}$$

The last series converges because $s > \rho$.

Part (ii) of the theorem is a noteworthy fact, which we shall use in a later part of this chapter.

We give two simple examples of the theorem; each of these shows that the condition $s > \rho$ cannot be improved.

EXAMPLE 1. Consider $f(z) = \sin \pi z$. Recall Euler's identity, namely

$$f(z) = \frac{e^{i\pi z} - e^{-i\pi z}}{2i},$$

which implies that $|f(z)| \leq e^{\pi|z|}$, and f has an order of growth ≤ 1 . By taking $z = ix$, where $x \in \mathbb{R}$, it is clear that the order of growth of f is actually equal to 1. However, f vanishes to order 1 at $z = n$ for each $n \in \mathbb{Z}$, and $\sum_{n \neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

EXAMPLE 2. Consider $f(z) = \cos z^{1/2}$, which we *define* by

$$\cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{(2n)!}.$$

Then f is entire, and it is easy to see that

$$|f(z)| \leq e^{|z|^{1/2}},$$

and the order of growth of f is $1/2$. Moreover, $f(z)$ vanishes when $z_n = ((n + 1/2)\pi)^2$, while $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

A natural question is whether or not, given any sequence of complex numbers $z_1, z_2, \dots$, there exists an entire function f with zeros precisely at the points of this sequence. A necessary condition is that $z_1, z_2, \dots$ do not accumulate, in other words we must have

$$\lim_{k \rightarrow \infty} |z_k| = \infty,$$

otherwise f would vanish identically by Theorem 4.8 in Chapter 2. Weierstrass proved that this condition is also sufficient by explicitly constructing a function with these prescribed zeros. A first guess is of course the product

$$(z - z_1)(z - z_2) \cdots,$$

which provides a solution in the special case when the sequence of zeros is finite. In general, Weierstrass showed how to insert factors in this product so that the convergence is guaranteed, yet no new zeros are introduced.

Before coming to the general construction, we review infinite products and study a basic example.

3 Infinite products

3.1 Generalities

Given a sequence $\{a_n\}_{n=1}^{\infty}$ of complex numbers, we say that the product

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if the limit

$$\lim_{N \rightarrow \infty} \prod_{n=1}^N (1 + a_n)$$

of the partial products exists.

A useful necessary condition that guarantees the existence of a product is contained in the following proposition.

Proposition 3.1 *If $\sum |a_n| < \infty$, then the product $\prod_{n=1}^{\infty} (1 + a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.*

This is simply Proposition 1.9 of Chapter 8 in Book I. We repeat the proof here.

Proof. If $\sum |a_n|$ converges, then for all large n we must have $|a_n| < 1/2$. Disregarding if necessary finitely many terms, we may assume that this inequality holds for all n . In particular, we can define $\log(1 + a_n)$ by the usual power series (see (6) in Chapter 3), and this logarithm satisfies the property that $1 + z = e^{\log(1+z)}$ whenever $|z| < 1$. Hence we may write the partial products as follows:

$$\prod_{n=1}^N (1 + a_n) = \prod_{n=1}^N e^{\log(1+a_n)} = e^{B_N},$$

where $B_N = \sum_{n=1}^N b_n$ with $b_n = \log(1 + a_n)$. By the power series expansion we see that $|\log(1 + z)| \leq 2|z|$, if $|z| < 1/2$. Hence $|b_n| \leq 2|a_n|$, so B_N converges as $N \rightarrow \infty$ to a complex number, say B . Since the exponential function is continuous, we conclude that e^{B_N} converges to e^B as $N \rightarrow \infty$, proving the first assertion of the proposition. Observe also that if $1 + a_n \neq 0$ for all n , then the product converges to a non-zero limit since it is expressed as e^B .

More generally, we can consider products of holomorphic functions.

Proposition 3.2 *Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω . If there exist constants $c_n > 0$ such that*

$$\sum c_n < \infty \quad \text{and} \quad |F_n(z) - 1| \leq c_n \quad \text{for all } z \in \Omega,$$

then:

- (i) *The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.*
- (ii) *If $F_n(z)$ does not vanish for any n , then*

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F'_n(z)}{F_n(z)}.$$

Proof. To prove the first statement, note that for each z we may argue as in the previous proposition if we write $F_n(z) = 1 + a_n(z)$, with

$|a_n(z)| \leq c_n$. Then, we observe that the estimates are actually uniform in z because the c_n 's are constants. It follows that the product converges uniformly to a holomorphic function, which we denote by $F(z)$.

To establish the second part of the theorem, suppose that K is a compact subset of Ω , and let

$$G_N(z) = \prod_{n=1}^N F_n(z).$$

We have just proved that $G_N \rightarrow F$ uniformly in Ω , so by Theorem 5.3 in Chapter 2, the sequence $\{G'_N\}$ converges uniformly to F' in K . Since G_N is uniformly bounded from below on K , we conclude that $G'_N/G_N \rightarrow F'/F$ uniformly on K , and because K is an arbitrary compact subset of Ω , the limit holds for every point of Ω . Moreover, as we saw in Section 4 of Chapter 3

$$\frac{G'_N}{G_N} = \sum_{n=1}^N \frac{F'_n}{F_n},$$

so part (ii) of the proposition is also proved.

3.2 Example: the product formula for the sine function

Before proceeding with the general theory of Weierstrass products, we consider the key example of the product formula for the sine function:

$$(3) \quad \frac{\sin \pi z}{\pi} = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

This identity will in turn be derived from the sum formula for the cotangent function ($\cot \pi z = \cos \pi z / \sin \pi z$):

$$(4) \quad \pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}.$$

The first formula holds for all complex numbers z , and the second whenever z is not an integer. The sum $\sum_{n=-\infty}^{\infty} 1/(z+n)$ needs to be properly understood, because the separate halves corresponding to positive and negative values of n do not converge. Only when interpreted symmetrically, as $\lim_{N \rightarrow \infty} \sum_{|n| \leq N} 1/(z+n)$, does the cancellation of terms lead to a convergent series as in (4) above.

We prove (4) by showing that both $\pi \cot \pi z$ and the series have the same structural properties. In fact, observe that if $F(z) = \pi \cot \pi z$, then F has the following three properties:

- (i) $F(z+1) = F(z)$ whenever z is not an integer.
- (ii) $F(z) = \frac{1}{z} + F_0(z)$, where F_0 is analytic near 0.
- (iii) $F(z)$ has simple poles at the integers, and no other singularities.

Then, we note that the function

$$\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n}$$

also satisfies these same three properties. In fact, property (i) is simply the observation that the passage from z to $z+1$ merely shifts the terms in the infinite sum. To be precise,

$$\sum_{|n| \leq N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n| \leq N} \frac{1}{z+n}.$$

Letting N tend to infinity proves the assertion. Properties (ii) and (iii) are evident from the representation $\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$ of the sum.

Therefore, the function defined by

$$\Delta(z) = F(z) - \sum_{n=-\infty}^{\infty} \frac{1}{z+n}$$

is periodic in the sense that $\Delta(z+1) = \Delta(z)$, and by (ii) the singularity of Δ at the origin is removable, and hence by periodicity the singularities at all the integers are also removable; this implies that Δ is entire.

To prove our formula, it will suffice to show that the function Δ is bounded in the complex plane. By the periodicity above, it is enough to do so in the strip $|\operatorname{Re}(z)| \leq 1/2$. This is because every $z' \in \mathbb{C}$ is of the form $z' = z + k$, where z is in the strip and k is an integer. Since Δ is holomorphic, it is bounded in the rectangle $|\operatorname{Im}(z)| \leq 1$, and we need only control the behavior of that function for $|\operatorname{Im}(z)| > 1$. If $\operatorname{Im}(z) > 1$ and $z = x + iy$, then

$$\cot \pi z = i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}},$$

and in absolute value this quantity is bounded. Also

$$\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} = \frac{1}{x + iy} + \sum_{n=1}^{\infty} \frac{2(x + iy)}{x^2 - y^2 - n^2 + 2ixy};$$

therefore if $y > 1$, we have

$$\left| \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} \right| \leq C + C \sum_{n=1}^{\infty} \frac{y}{y^2 + n^2}.$$

Now the sum on the right-hand side is majorized by

$$\int_0^{\infty} \frac{y}{y^2 + x^2} dx,$$

because the function $y/(y^2 + x^2)$ is decreasing in x ; moreover, as the change of variables $x \mapsto yx$ shows, the integral is independent of y , and hence bounded. By a similar argument Δ is bounded in the strip where $\operatorname{Im}(z) < -1$, hence is bounded throughout the whole strip $|\operatorname{Re}(z)| \leq 1/2$. Therefore Δ is bounded in $\mathbb{C}$, and by Liouville's theorem, $\Delta(z)$ is constant. The observation that Δ is odd shows that this constant must be 0, and concludes the proof of formula (4).

To prove (3), we now let

$$G(z) = \frac{\sin \pi z}{\pi} \quad \text{and} \quad P(z) = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2} \right).$$

Proposition 3.2 and the fact that $\sum 1/n^2 < \infty$ guarantee that the product $P(z)$ converges, and that away from the integers we have

$$\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}.$$

Since $G'(z)/G(z) = \pi \cot \pi z$, the cotangent formula (4) gives

$$\left(\frac{P(z)}{G(z)} \right)' = \frac{P(z)}{G(z)} \left[\frac{P'(z)}{P(z)} - \frac{G'(z)}{G(z)} \right] = 0,$$

and so $P(z) = cG(z)$ for some constant c . Dividing this identity by z , and taking the limit as $z \rightarrow 0$, we find $c = 1$.

Remark. Other proofs of (4) and (3) can be given by integrating analogous identities for $\pi^2/(\sin \pi z)^2$ derived in Exercise 12, Chapter 3, and Exercise 7, Chapter 4. Still other proofs using Fourier series can be found in the exercises of Chapters 3 and 5 of Book I.

4 Weierstrass infinite products

We now turn to Weierstrass's construction of an entire function with prescribed zeros.

Theorem 4.1 *Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \rightarrow \infty$ as $n \rightarrow \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.*

Recall that if a holomorphic function f vanishes at $z = a$, then the multiplicity of the zero a is the integer m so that

$$f(z) = (z - a)^m g(z),$$

where g is holomorphic and nowhere vanishing in a neighborhood of a . Alternatively, m is the first non-zero power of $z - a$ in the power series expansion of f at a . Since, as before, we allow for repetitions in the sequence $\{a_n\}$, the theorem actually guarantees the existence of entire functions with prescribed zeros and with desired multiplicities.

To begin the proof, note first that if f_1 and f_2 are two entire functions that vanish at all $z = a_n$ and nowhere else, then f_1/f_2 has removable singularities at all the points a_n . Hence f_1/f_2 is entire and vanishes nowhere, so that there exists an entire function g with $f_1(z)/f_2(z) = e^{g(z)}$, as we showed in Section 6 of Chapter 3. Therefore $f_1(z) = f_2(z)e^{g(z)}$ and the last statement of the theorem is verified.

Hence we are left with the task of constructing a function that vanishes at all the points of the sequence $\{a_n\}$ and nowhere else. A naive guess, suggested by the product formula for $\sin \pi z$, is the product $\prod_n (1 - z/a_n)$. The problem is that this product converges only for suitable sequences $\{a_n\}$, so we correct this by inserting exponential factors. These factors will make the product converge without adding new zeros.

For each integer $k \geq 0$ we define **canonical factors** by

$$E_0(z) = 1 - z \quad \text{and} \quad E_k(z) = (1 - z)e^{z + z^2/2 + \cdots + z^k/k}, \quad \text{for } k \geq 1.$$

The integer k is called the **degree** of the canonical factor.

Lemma 4.2 *If $|z| \leq 1/2$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$.*

Proof. If $|z| \leq 1/2$, then with the logarithm defined in terms of the power series, we have $1 - z = e^{\log(1-z)}$, and therefore

$$E_k(z) = e^{\log(1-z) + z + z^2/2 + \cdots + z^k/k} = e^w,$$

where $w = -\sum_{n=k+1}^{\infty} z^n/n$. Observe that since $|z| \leq 1/2$ we have

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \leq 2|z|^{k+1}.$$

In particular, we have $|w| \leq 1$ and this implies that

$$|1 - E_k(z)| = |1 - e^w| \leq c'|w| \leq c|z|^{k+1}.$$

Remark. An important technical point is that the constant c in the statement of the lemma can be chosen to be independent of k . In fact, an examination of the proof shows that we may take $c' = e$ and then $c = 2e$.

Suppose that we are given a zero of order m at the origin, and that $a_1, a_2, \dots$ are all non-zero. Then we define the Weierstrass product by

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

We claim that this function has the required properties; that is, f is entire with a zero of order m at the origin, zeros at each point of the sequence $\{a_n\}$, and f vanishes nowhere else.

Fix $R > 0$, and suppose that z belongs to the disc $|z| < R$. We shall prove that f has all the desired properties in this disc, and since R is arbitrary, this will prove the theorem.

We can consider two types of factors in the formula defining f , with the choice depending on whether $|a_n| \leq 2R$ or $|a_n| > 2R$. There are only finitely many terms of the first kind (since $|a_n| \rightarrow \infty$), and we see that the finite product vanishes at all $z = a_n$ with $|a_n| < R$. If $|a_n| \geq 2R$, we have $|z/a_n| \leq 1/2$, hence the previous lemma implies

$$|1 - E_n(z/a_n)| \leq c \left| \frac{z}{a_n} \right|^{n+1} \leq \frac{c}{2^{n+1}}.$$

Note that by the above remark, c does not depend on n . Therefore, the product

$$\prod_{|a_n| \geq 2R} E_n(z/a_n)$$

defines a holomorphic function when $|z| < R$, and does not vanish in that disc by the propositions in Section 3. This shows that the function

f has the desired properties, and the proof of Weierstrass's theorem is complete.

5 Hadamard's factorization theorem

The theorem of this section combines the results relating the growth of a function to the number of zeros it possesses, and the above product theorem. Weierstrass's theorem states that a function that vanishes at the points $a_1, a_2, \dots$ takes the form

$$e^{g(z)} z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Hadamard refined this result by showing that in the case of functions of finite order, the degree of the canonical factors can be taken to be constant, and g is then a polynomial.

Recall that an entire function has an order of growth $\leq \rho$ if

$$|f(z)| \leq A e^{B|z|^\rho},$$

and that *the* order of growth ρ_0 of f is the infimum of all such ρ 's.

A basic result we proved earlier was that if f has order of growth $\leq \rho$, then

$$n(r) \leq Cr^\rho, \quad \text{for all large } r,$$

and if $a_1, a_2, \dots$ are the non-zero zeros of f , and $s > \rho$, then

$$\sum |a_n|^{-s} < \infty.$$

Theorem 5.1 *Suppose f is entire and has growth order ρ_0 . Let k be the integer so that $k \leq \rho_0 < k+1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f , then*

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\leq k$, and m is the order of the zero of f at $z = 0$.

Main lemmas

Here we gather a few lemmas needed in the proof of Hadamard's theorem.

Lemma 5.2 *The canonical products satisfy*

$$|E_k(z)| \geq e^{-c|z|^{k+1}} \quad \text{if } |z| \leq 1/2$$

and

$$|E_k(z)| \geq |1 - z| e^{-c'|z|^k} \quad \text{if } |z| \geq 1/2.$$

Proof. If $|z| \leq 1/2$ we can use the power series to define the logarithm of $1 - z$, so that

$$E_k(z) = e^{\log(1-z) + \sum_{n=1}^k z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w.$$

Since $|e^w| \geq e^{-|w|}$ and $|w| \leq c|z|^{k+1}$, the first part of the lemma follows. For the second part, simply observe that if $|z| \geq 1/2$, then

$$|E_k(z)| = |1 - z| |e^{z+z^2/2+\dots+z^k/k}|,$$

and that there exists $c' > 0$ such that

$$|e^{z+z^2/2+\dots+z^k/k}| \geq e^{-|z+z^2/2+\dots+z^k/k|} \geq e^{-c'|z|^k}.$$

The inequality in the lemma when $|z| \geq 1/2$ then follows from these observations.

The key to the proof of Hadamard's theorem consists of finding a lower bound for the product of the canonical factors when z stays away from the zeros $\{a_n\}$. Therefore, we shall first estimate the product from below, in the complement of small discs centered at these points.

Lemma 5.3 *For any s with $\rho_0 < s < k + 1$, we have*

$$\left| \prod_{n=1}^{\infty} E_k(z/a_n) \right| \geq e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \dots$

Proof. The proof this lemma is a little subtle. First, we write

$$\prod_{n=1}^{\infty} E_k(z/a_n) = \prod_{|a_n| \leq 2|z|} E_k(z/a_n) \prod_{|a_n| > 2|z|} E_k(z/a_n).$$

For the second product the estimate asserted above holds with no restriction on z . Indeed, by the previous lemma

$$\begin{aligned} \left| \prod_{|a_n| > 2|z|} E_k(z/a_n) \right| &= \prod_{|a_n| > 2|z|} |E_k(z/a_n)| \\ &\geq \prod_{|a_n| > 2|z|} e^{-c|z/a_n|^{k+1}} \\ &\geq e^{-c|z|^{k+1} \sum_{|a_n| > 2|z|} |a_n|^{-k-1}}. \end{aligned}$$

But $|a_n| > 2|z|$ and $s < k + 1$, so we must have

$$|a_n|^{-k-1} = |a_n|^{-s} |a_n|^{s-k-1} \leq C |a_n|^{-s} |z|^{s-k-1}.$$

Therefore, the fact that $\sum |a_n|^{-s}$ converges implies that

$$\left| \prod_{|a_n| > 2|z|} E_k(z/a_n) \right| \geq e^{-c|z|^s}$$

for some $c > 0$.

To estimate the first product, we use the second part of Lemma 5.2, and write

$$(5) \quad \left| \prod_{|a_n| \leq 2|z|} E_k(z/a_n) \right| \geq \prod_{|a_n| \leq 2|z|} \left| 1 - \frac{z}{a_n} \right| \prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k}.$$

We now note that

$$\prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n| \leq 2|z|} |a_n|^{-k}},$$

and again, we have $|a_n|^{-k} = |a_n|^{-s} |a_n|^{s-k} \leq C |a_n|^{-s} |z|^{s-k}$, thereby proving that

$$\prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k} \geq e^{-c|z|^s}.$$

It is the estimate on the first product on the right-hand side of (5) which requires the restriction on z imposed in the statement of the

lemma. Indeed, whenever z does not belong to a disc of radius $|a_n|^{-k-1}$ centered at a_n , we must have $|a_n - z| \geq |a_n|^{-k-1}$. Therefore

$$\begin{aligned} \prod_{|a_n| \leq 2|z|} \left| 1 - \frac{z}{a_n} \right| &= \prod_{|a_n| \leq 2|z|} \left| \frac{a_n - z}{a_n} \right| \\ &\geq \prod_{|a_n| \leq 2|z|} |a_n|^{-k-1} |a_n|^{-1} \\ &= \prod_{|a_n| \leq 2|z|} |a_n|^{-k-2}. \end{aligned}$$

Finally, the estimate for the first product follows from the fact that

$$\begin{aligned} (k+2) \sum_{|a_n| \leq 2|z|} \log |a_n| &\leq (k+2) \mathfrak{n}(2|z|) \log 2|z| \\ &\leq c|z|^s \log 2|z| \\ &\leq c'|z|^{s'}, \end{aligned}$$

for any $s' > s$, and the second inequality follows because $\mathfrak{n}(2|z|) \leq c|z|^s$ by Theorem 2.1. Since we restricted s to satisfy $s > \rho_0$, we can take an initial s sufficiently close to ρ_0 , so that the assertion of the lemma is established (with s being replaced by s').

Corollary 5.4 *There exists a sequence of radii, $r_1, r_2, \dots$, with $r_m \rightarrow \infty$, such that*

$$\left| \prod_{n=1}^{\infty} E_k(z/a_n) \right| \geq e^{-c|z|^s} \quad \text{for } |z| = r_m.$$

Proof. Since $\sum |a_n|^{-k-1} < \infty$, there exists an integer N so that

$$\sum_{n=N}^{\infty} |a_n|^{-k-1} < 1/10.$$

Therefore, given any two consecutive large integers L and $L+1$, we can find a positive number r with $L \leq r \leq L+1$, such that the circle of radius r centered at the origin does not intersect the forbidden discs of Lemma 5.3. For otherwise, the union of the intervals

$$I_n = \left[|a_n| - \frac{1}{|a_n|^{k+1}}, |a_n| + \frac{1}{|a_n|^{k+1}} \right]$$

(which are of length $2|a_n|^{-k-1}$) would cover all the interval $[L, L+1]$. (See Figure 1.) This would imply $2 \sum_{n=N}^{\infty} |a_n|^{-k-1} \geq 1$, which is a contradiction. We can then apply the previous lemma with $|z| = r$ to conclude the proof of the corollary.

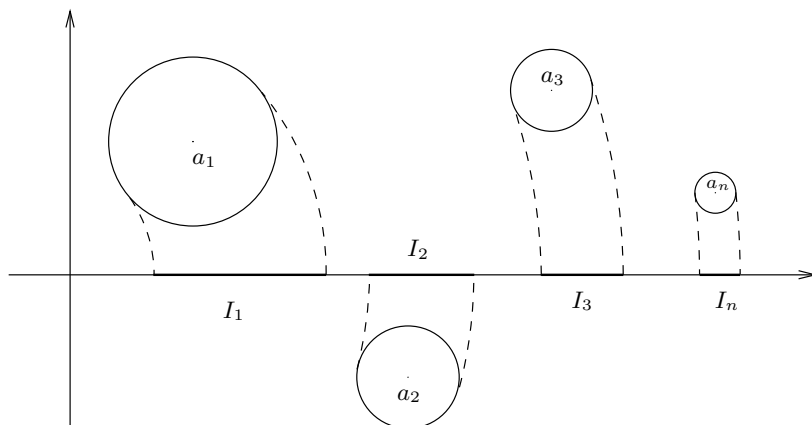


Figure 1. The intervals I_n

Proof of Hadamard's theorem

Let

$$E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n).$$

To prove that E is entire, we repeat the argument in the proof of Theorem 4.1; we take into account that by Lemma 4.2

$$|1 - E_k(z/a_n)| \leq c \left| \frac{z}{a_n} \right|^{k+1} \quad \text{for all large } n,$$

and that the series $\sum |a_n|^{-k-1}$ converges. (Recall $\rho_0 < s < k+1$.) Moreover, E has the zeros of f , therefore f/E is holomorphic and nowhere vanishing. Hence

$$\frac{f(z)}{E(z)} = e^{g(z)}$$

for some entire function g . By the fact that f has growth order ρ_0 , and because of the estimate from below for E obtained in Corollary 5.4, we have

$$e^{\operatorname{Re}(g(z))} = \left| \frac{f(z)}{E(z)} \right| \leq c' e^{c|z|^s}$$

for $|z| = r_m$. This proves that

$$\operatorname{Re}(g(z)) \leq C|z|^s, \quad \text{for } |z| = r_m.$$

The proof of Hadamard's theorem is therefore complete if we can establish the following final lemma.

Lemma 5.5 *Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies*

$$u(z) \leq Cr^s \quad \text{whenever } |z| = r$$

for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\leq s$.

Proof. We can expand g in a power series centered at the origin

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

We have already proved in the last section of Chapter 3 (as a simple application of Cauchy's integral formulas) that

$$(6) \quad \frac{1}{2\pi} \int_0^{2\pi} g(re^{i\theta}) e^{-in\theta} d\theta = \begin{cases} a_n r^n & \text{if } n \geq 0 \\ 0 & \text{if } n < 0. \end{cases}$$

By taking complex conjugates we find that

$$(7) \quad \frac{1}{2\pi} \int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta} d\theta = 0$$

whenever $n > 0$, and since $2u = g + \overline{g}$ we add equations (6) and (7) to obtain

$$a_n r^n = \frac{1}{\pi} \int_0^{2\pi} u(re^{i\theta}) e^{-in\theta} d\theta, \quad \text{whenever } n > 0.$$

For $n = 0$ we can simply take real parts of both sides of (6) to find that

$$2\operatorname{Re}(a_0) = \frac{1}{\pi} \int_0^{2\pi} u(re^{i\theta}) d\theta.$$

Now we recall the simple fact that whenever $n \neq 0$, the integral of $e^{-in\theta}$ over any circle centered at the origin vanishes. Therefore

$$a_n = \frac{1}{\pi r^n} \int_0^{2\pi} [u(re^{i\theta}) - Cr^s] e^{-in\theta} d\theta \quad \text{when } n > 0,$$

hence

$$|a_n| \leq \frac{1}{\pi r^n} \int_0^{2\pi} [Cr^s - u(re^{i\theta})] d\theta \leq 2Cr^{s-n} - 2\operatorname{Re}(a_0)r^{-n}.$$

Letting r tend to infinity along the sequence given in the hypothesis of the lemma proves that $a_n = 0$ for $n > s$. This completes the proof of the lemma and of Hadamard's theorem.

6 Exercises

1. Give another proof of Jensen's formula in the unit disc using the functions (called Blaschke factors)

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

[Hint: The function $f/(\psi_{z_1} \cdots \psi_{z_N})$ is nowhere vanishing.]

2. Find the order of growth of the following entire functions:

(a) $p(z)$ where p is a polynomial.

(b) e^{bz^n} for $b \neq 0$.

(c) e^{e^z} .

3. Show that if τ is fixed with $\operatorname{Im}(\tau) > 0$, then the Jacobi theta function

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

is of order 2 as a function of z . Further properties of Θ will be studied in Chapter 10.

[Hint: $-n^2 t + 2n|z| \leq -n^2 t/2$ when $t > 0$ and $n \geq 4|z|/t$.]

4. Let $t > 0$ be given and fixed, and define $F(z)$ by

$$F(z) = \prod_{n=1}^{\infty} (1 - e^{-2\pi n t} e^{2\pi i z}).$$

Note that the product defines an entire function of z .

- (a) Show that $|F(z)| \leq Ae^{a|z|^2}$, hence F is of order 2.
- (b) F vanishes exactly when $z = -int + m$ for $n \geq 1$ and n, m integers. Thus, if z_n is an enumeration of these zeros we have

$$\sum \frac{1}{|z_n|^2} = \infty \quad \text{but} \quad \sum \frac{1}{|z_n|^{2+\epsilon}} < \infty.$$

[Hint: To prove (a), write $F(z) = F_1(z)F_2(z)$ where

$$F_1(z) = \prod_{n=1}^N (1 - e^{-2\pi nt} e^{2\pi iz}) \quad \text{and} \quad F_2(z) = \prod_{n=N+1}^{\infty} (1 - e^{-2\pi nt} e^{2\pi iz}).$$

Choose $N \approx c|z|$ with c appropriately large. Then, since

$$\left(\sum_{N+1}^{\infty} e^{-2\pi nt} \right) e^{2\pi|z|} \leq 1,$$

one has $|F_2(z)| \leq A$. However,

$$|1 - e^{-2\pi nt} e^{2\pi iz}| \leq 1 + e^{2\pi|z|} \leq 2e^{2\pi|z|}.$$

Thus $|F_1(z)| \leq 2^N e^{2\pi N|z|} \leq e^{c'|z|^2}$. Note that a simple variant of the function F arises as a factor in the triple product formula for the Jacobi theta function Θ , taken up in Chapter 10.]

5. Show that if $\alpha > 1$, then

$$F_{\alpha}(z) = \int_{-\infty}^{\infty} e^{-|t|^{\alpha}} e^{2\pi izt} dt$$

is an entire function of growth order $\alpha/(\alpha - 1)$.

[Hint: Show that

$$-\frac{|t|^{\alpha}}{2} + 2\pi|z||t| \leq c|z|^{\alpha/(\alpha-1)}$$

by considering the two cases $|t|^{\alpha-1} \leq A|z|$ and $|t|^{\alpha-1} \geq A|z|$, for an appropriate constant A .]

6. Prove Wallis's product formula

$$\frac{\pi}{2} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots \frac{2m \cdot 2m}{(2m-1) \cdot (2m+1)} \cdots$$

[Hint: Use the product formula for $\sin z$ at $z = \pi/2$.]

7. Establish the following properties of infinite products.

- (a) Show that if $\sum |a_n|^2$ converges, then the product $\prod(1 + a_n)$ converges to a non-zero limit if and only if $\sum a_n$ converges.
- (b) Find an example of a sequence of complex numbers $\{a_n\}$ such that $\sum a_n$ converges but $\prod(1 + a_n)$ diverges.
- (c) Also find an example such that $\prod(1 + a_n)$ converges and $\sum a_n$ diverges.

8. Prove that for every z the product below converges, and

$$\cos(z/2) \cos(z/4) \cos(z/8) \cdots = \prod_{k=1}^{\infty} \cos(z/2^k) = \frac{\sin z}{z}.$$

[Hint: Use the fact that $\sin 2z = 2 \sin z \cos z$.]

9. Prove that if $|z| < 1$, then

$$(1+z)(1+z^2)(1+z^4)(1+z^8) \cdots = \prod_{k=0}^{\infty} (1+z^{2^k}) = \frac{1}{1-z}.$$

10. Find the Hadamard products for:

(a) $e^z - 1$;

(b) $\cos \pi z$.

[Hint: The answers are $e^{z/2} z \prod_{n=1}^{\infty} (1 + z^2/4n^2\pi^2)$ and $\prod_{n=0}^{\infty} (1 - 4z^2/(2n+1)^2)$, respectively.]

11. Show that if f is an entire function of finite order that omits two values, then f is constant. This result remains true for any entire function and is known as Picard's little theorem.

[Hint: If f misses a , then $f(z) - a$ is of the form $e^{p(z)}$ where p is a polynomial.]

12. Suppose f is entire and never vanishes, and that none of the higher derivatives of f ever vanish. Prove that if f is also of finite order, then $f(z) = e^{az+b}$ for some constants a and b .

13. Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.

[Hint: Apply Hadamard's theorem.]

14. Deduce from Hadamard's theorem that if F is entire and of growth order ρ that is non-integral, then F has infinitely many zeros.

15. Prove that every meromorphic function in $\mathbb{C}$ is the quotient of two entire functions. Also, if $\{a_n\}$ and $\{b_n\}$ are two disjoint sequences having no finite limit

points, then there exists a meromorphic function in the whole complex plane that vanishes exactly at $\{a_n\}$ and has poles exactly at $\{b_n\}$.

16. Suppose that

$$Q_n(z) = \sum_{k=1}^{N_n} c_k^n z^k$$

are given polynomials for $n = 1, 2, \dots$. Suppose also that we are given a sequence of complex numbers $\{a_n\}$ without limit points. Prove that there exists a meromorphic function $f(z)$ whose only poles are at $\{a_n\}$, and so that for each n , the difference

$$f(z) - Q_n\left(\frac{1}{z - a_n}\right)$$

is holomorphic near a_n . In other words, f has a prescribed poles and principal parts at each of these poles. This result is due to Mittag-Leffler.

17. Given two countably infinite sequences of complex numbers $\{a_k\}_{k=0}^\infty$ and $\{b_k\}_{k=0}^\infty$, with $\lim_{k \rightarrow \infty} |a_k| = \infty$, it is always possible to find an entire function F that satisfies $F(a_k) = b_k$ for all k .

- (a) Given n distinct complex numbers $a_1, \dots, a_n$, and another n complex numbers $b_1, \dots, b_n$, construct a polynomial P of degree $\leq n - 1$ with

$$P(a_i) = b_i \quad \text{for } i = 1, \dots, n.$$

- (b) Let $\{a_k\}_{k=0}^\infty$ be a sequence of distinct complex numbers such that $a_0 = 0$ and $\lim_{k \rightarrow \infty} |a_k| = \infty$, and let $E(z)$ denote a Weierstrass product associated with $\{a_k\}$. Given complex numbers $\{b_k\}_{k=0}^\infty$, show that there exist integers $m_k \geq 1$ such that the series

$$F(z) = \frac{b_0}{E'(z)} \frac{E(z)}{z} + \sum_{k=1}^{\infty} \frac{b_k}{E'(a_k)} \frac{E(z)}{z - a_k} \left(\frac{z}{a_k}\right)^{m_k}$$

defines an entire function that satisfies

$$F(a_k) = b_k \quad \text{for all } k \geq 0.$$

This is known as the Pringsheim interpolation formula.

7 Problems

1. Prove that if f is holomorphic in the unit disc, bounded and not identically zero, and $z_1, z_2, \dots, z_n, \dots$ are its zeros ($|z_k| < 1$), then

$$\sum_n (1 - |z_n|) < \infty.$$

[Hint: Use Jensen's formula.]

2.* In this problem, we discuss Blaschke products, which are bounded analogues in the disc of the Weierstrass products for entire functions.

(a) Show that for $0 < |\alpha| < 1$ and $|z| \leq r < 1$ the inequality

$$\left| \frac{\alpha + |\alpha|z}{(1 - \bar{\alpha}z)\alpha} \right| \leq \frac{1+r}{1-r}$$

holds.

(b) Let $\{\alpha_n\}$ be a sequence in the unit disc such that $\alpha_n \neq 0$ for all n and

$$\sum_{n=1}^{\infty} (1 - |\alpha_n|) < \infty.$$

Note that this will be the case if $\{\alpha_n\}$ are the zeros of a bounded holomorphic function on the unit disc (see Problem 1). Show that the product

$$f(z) = \prod_{n=1}^{\infty} \frac{\alpha_n - z}{1 - \bar{\alpha}_n z} \frac{|\alpha_n|}{\alpha_n}$$

converges uniformly for $|z| \leq r < 1$, and defines a holomorphic function on the unit disc having precisely the zeros α_n and no other zeros. Show that $|f(z)| \leq 1$.

3.* Show that $\sum \frac{z^n}{(n!)^\alpha}$ is an entire function of order $1/\alpha$.

4.* Let $F(z) = \sum_{n=0}^{\infty} a_n z^n$ be an entire function of finite order. Then the growth order of F is intimately linked with the growth of the coefficients a_n as $n \rightarrow \infty$. In fact:

(a) Suppose $|F(z)| \leq A e^{a|z|^\rho}$. Then

$$(8) \quad \limsup_{n \rightarrow \infty} |a_n|^{1/n} n^{1/\rho} < \infty.$$

(b) Conversely, if (8) holds, then $|F(z)| \leq A_\epsilon e^{a_\epsilon |z|^{\rho+\epsilon}}$, for every $\epsilon > 0$.

[Hint: To prove (a), use Cauchy's inequality

$$|a_n| \leq \frac{A}{r^n} e^{ar^\rho},$$

and the fact that the function $u^{-n} e^{u^\rho}$, $0 < u < \rho$, attains its minimum value $e^{n/\rho} (\rho/n)^{n/\rho}$ at $u = n^{1/\rho} / \rho^{1/\rho}$. Then, choose r in terms of n to achieve this minimum.

To establish (b), note that for $|z| = r$,

$$|F(z)| \leq \sum \frac{c^n r^n}{n^{n/\rho}} \leq \sum \frac{c^n r^n}{(n!)^{1/\rho}}$$

for some constant c , since $n^n \geq n!$. This yields a reduction to Problem 3.]

6 The Gamma and Zeta Functions

It is no exaggeration to say that the gamma and zeta functions are among the most important nonelementary functions in mathematics. The gamma function Γ is ubiquitous in nature. It arises in a host of calculations and is featured in a large number of identities that occur in analysis. Part of the explanation for this probably lies in the basic structural property of the gamma function, which essentially characterizes it: $1/\Gamma(s)$ is the (simplest) entire function¹ which has zeros at exactly $s = 0, -1, -2, \dots$

The zeta function ζ (whose study, like that of the gamma function, was initiated by Euler) plays a fundamental role in the analytic theory of numbers. Its intimate connection with prime numbers comes about via the identity for $\zeta(s)$:

$$\prod_p \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

where the product is over all primes. The behavior of $\zeta(s)$ for real $s > 1$, with s tending to 1, was used by Euler to prove that $\sum_p 1/p$ diverges, and a similar reasoning for L -functions is at the starting point of the proof of Dirichlet's theorem on primes in arithmetic progression, as we saw in Book I.

While there is no difficulty in seeing that $\zeta(s)$ is well-defined (and analytic) when $\operatorname{Re}(s) > 1$, it was Riemann who realized that the further study of primes was bound up with the analytic (in fact, meromorphic) continuation of ζ into the rest of the complex plane. Beyond this, we also consider its remarkable functional equation, which reveals a symmetry about the line $\operatorname{Re}(s) = 1/2$, and whose proof is based on a corresponding identity for the theta function. We also make a more detailed study of the growth of $\zeta(s)$ near the line $\operatorname{Re}(s) = 1$, which will be required in the proof of the prime number theorem given in the next chapter.

¹In keeping with the standard notation of the subject, we denote by s (instead of z) the argument of the functions Γ and ζ .

1 The gamma function

For $s > 0$, the **gamma function** is defined by

$$(1) \quad \Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt.$$

The integral converges for each positive s because near $t = 0$ the function t^{s-1} is integrable, and for t large the convergence is guaranteed by the exponential decay of the integrand. These observations allow us to extend the domain of definition of Γ as follows.

Proposition 1.1 *The gamma function extends to an analytic function in the half-plane $\operatorname{Re}(s) > 0$, and is still given there by the integral formula (1).*

Proof. It suffices to show that the integral defines a holomorphic function in every strip

$$S_{\delta, M} = \{\delta < \operatorname{Re}(s) < M\},$$

where $0 < \delta < M < \infty$. Note that if σ denotes the real part of s , then $|e^{-t} t^{s-1}| = e^{-t} t^{\sigma-1}$, so that the integral

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt,$$

which is defined by the limit $\lim_{\epsilon \rightarrow 0} \int_\epsilon^{1/\epsilon} e^{-t} t^{s-1} dt$, converges for each $s \in S_{\delta, M}$. For $\epsilon > 0$, let

$$F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t} t^{s-1} dt.$$

By Theorem 5.4 in Chapter 2, the function F_ϵ is holomorphic in the strip $S_{\delta, M}$. By Theorem 5.2, also of Chapter 2, it suffices to show that F_ϵ converges uniformly to Γ on the strip $S_{\delta, M}$. To see this, we first observe that

$$|\Gamma(s) - F_\epsilon(s)| \leq \int_0^\epsilon e^{-t} t^{\sigma-1} dt + \int_{1/\epsilon}^\infty e^{-t} t^{\sigma-1} dt.$$

The first integral converges uniformly to 0, as ϵ tends to 0 since it can be easily estimated by ϵ^δ/δ whenever $0 < \epsilon < 1$. The second integral converges uniformly to 0 as well, since

$$\left| \int_{1/\epsilon}^\infty e^{-t} t^{\sigma-1} dt \right| \leq \int_{1/\epsilon}^\infty e^{-t} t^{M-1} dt \leq C \int_{1/\epsilon}^\infty e^{-t/2} dt \rightarrow 0,$$

and the proof is complete.

1.1 Analytic continuation

Despite the fact that the integral defining Γ is not absolutely convergent for other values of s , we can go further and prove that there exists a meromorphic function defined on all of $\mathbb{C}$ that equals Γ in the half-plane $\operatorname{Re}(s) > 0$. In the same sense as in Chapter 2, we say that this function is the analytic continuation² of Γ , and we therefore continue to denote it by Γ .

To prove the asserted analytic extension to a meromorphic function, we need a lemma, which incidentally exhibits an important property of Γ .

Lemma 1.2 *If $\operatorname{Re}(s) > 0$, then*

$$(2) \quad \Gamma(s+1) = s\Gamma(s).$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \dots$

Proof. Integrating by parts in the finite integrals gives

$$\int_{\epsilon}^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s) dt = - \int_{\epsilon}^{1/\epsilon} e^{-t}t^s dt + s \int_{\epsilon}^{1/\epsilon} e^{-t}t^{s-1} dt,$$

and the desired formula (2) follows by letting ϵ tend to 0, and noting that the left-hand side vanishes because $e^{-t}t^s \rightarrow 0$ as t tends to 0 or ∞ . Now it suffices to check that

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1,$$

and to apply (2) successively to find that $\Gamma(n+1) = n!$.

Formula (2) in the lemma is all we need to give a proof of the following theorem.

Theorem 1.3 *The function $\Gamma(s)$ initially defined for $\operatorname{Re}(s) > 0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the negative integers $s = 0, -1, \dots$. The residue of Γ at $s = -n$ is $(-1)^n/n!$.*

²Uniqueness of the analytic continuation is guaranteed since the complement of the poles of a meromorphic function forms a connected set.

Proof. It suffices to extend Γ to each half-plane $\operatorname{Re}(s) > -m$, where $m \geq 1$ is an integer. For $\operatorname{Re}(s) > -1$, we define

$$F_1(s) = \frac{\Gamma(s+1)}{s}.$$

Since $\Gamma(s+1)$ is holomorphic in $\operatorname{Re}(s) > -1$, we see that F_1 is meromorphic in that half-plane, with the only possible singularity a simple pole at $s = 0$. The fact that $\Gamma(1) = 1$ shows that F_1 does in fact have a simple pole at $s = 0$ with residue 1. Moreover, if $\operatorname{Re}(s) > 0$, then

$$F_1(s) = \frac{\Gamma(s+1)}{s} = \Gamma(s)$$

by the previous lemma. So F_1 extends Γ to a meromorphic function on the half-plane $\operatorname{Re}(s) > -1$. We can now continue in this fashion by defining a meromorphic F_m for $\operatorname{Re}(s) > -m$ that agrees with Γ on $\operatorname{Re}(s) > 0$. For $\operatorname{Re}(s) > -m$, where m is an integer ≥ 1 , define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

The function F_m is meromorphic in $\operatorname{Re}(s) > -m$ and has simple poles at $s = 0, -1, -2, \dots, -m+1$ with residues

$$\begin{aligned} \operatorname{res}_{s=-n} F_m(s) &= \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} \\ &= \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} \\ &= \frac{(-1)^n}{n!}. \end{aligned}$$

Successive applications of the lemma show that $F_m(s) = \Gamma(s)$ for $\operatorname{Re}(s) > 0$. By uniqueness, this also means that $F_m = F_k$ for $1 \leq k \leq m$ on the domain of definition of F_k . Therefore, we have obtained the desired continuation of Γ .

Remark. We have already proved that $\Gamma(s+1) = s\Gamma(s)$ whenever $\operatorname{Re}(s) > 0$. In fact, by analytic continuation, this formula remains true whenever $s \neq 0, -1, -2, \dots$, that is, whenever s is not a pole of Γ . This is because both sides of the formula are holomorphic in the complement of the poles of Γ and are equal when $\operatorname{Re}(s) > 0$. Actually, one can go further, and note that if s is a negative integer $s = -n$ with $n \geq 1$, then both sides of the formula are infinite and moreover

$$\operatorname{res}_{s=-n} \Gamma(s+1) = -n \operatorname{res}_{s=-n} \Gamma(s).$$

Finally, note that when $s = 0$ we have $\Gamma(1) = \lim_{s \rightarrow 0} s\Gamma(s)$.

An alternate proof of Theorem 1.3, which is interesting in its own right and whose ideas recur later, is obtained by splitting the integral for $\Gamma(s)$ defined on $\operatorname{Re}(s) > 0$ as follows:

$$\Gamma(s) = \int_0^1 e^{-t} t^{s-1} dt + \int_1^\infty e^{-t} t^{s-1} dt.$$

The integral on the far right defines an entire function; also expanding e^{-t} in a power series and integrating term by term gives

$$\int_0^1 e^{-t} t^{s-1} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)}.$$

Therefore

$$(3) \quad \Gamma(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t} t^{s-1} dt \quad \text{for } \operatorname{Re}(s) > 0.$$

Finally, the series defines a meromorphic function on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $s = -n$. To prove this, we argue as follows. For a fixed $R > 0$ we may split the sum in two parts

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} = \sum_{n=0}^N \frac{(-1)^n}{n!(n+s)} + \sum_{n=N+1}^{\infty} \frac{(-1)^n}{n!(n+s)},$$

where N is an integer chosen so that $N > 2R$. The first sum, which is finite, defines a meromorphic function in the disc $|s| < R$ with poles at the desired points and the correct residues. The second sum converges uniformly in that disc, hence defines a holomorphic function there, since $n > N > 2R$ and $|n+s| \geq R$ imply

$$\left| \frac{(-1)^n}{n!(n+s)} \right| \leq \frac{1}{n!R}.$$

Since R was arbitrary, we conclude that the series in (3) has the desired properties.

In particular, the relation (3) now holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

The following identity reveals the symmetry of Γ about the line $\operatorname{Re}(s) = 1/2$.

Theorem 1.4 For all $s \in \mathbb{C}$,

$$(4) \quad \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}.$$

Observe that $\Gamma(1-s)$ has simple poles at the positive integers $s = 1, 2, 3, \dots$, so that $\Gamma(s)\Gamma(1-s)$ is a meromorphic function on $\mathbb{C}$ with simple poles at *all* the integers, a property also shared by $\pi/\sin \pi s$.

To prove the identity, it suffices to do so for $0 < s < 1$ since it then holds on all of $\mathbb{C}$ by analytic continuation.

Lemma 1.5 For $0 < a < 1$, $\int_0^\infty \frac{v^{a-1}}{1+v} dv = \frac{\pi}{\sin \pi a}$.

Proof. We observe first that

$$\int_0^\infty \frac{v^{a-1}}{1+v} dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x} dx,$$

which follows by making the change of variables $v = e^x$. However, using contour integration, we saw in Example 2 of Section 2.1 in Chapter 3, that the second integral equals $\pi/\sin \pi a$, as desired.

To establish the theorem, we first note that for $0 < s < 1$ we may write

$$\Gamma(1-s) = \int_0^\infty e^{-u} u^{-s} du = t \int_0^\infty e^{-vt} (vt)^{-s} dv,$$

where for $t > 0$ we made the change of variables $vt = u$. This trick then gives

$$\begin{aligned} \Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t} t^{s-1} \Gamma(1-s) dt \\ &= \int_0^\infty e^{-t} t^{s-1} \left(t \int_0^\infty e^{-vt} (vt)^{-s} dv \right) dt \\ &= \int_0^\infty \int_0^\infty e^{-t[1+v]} v^{-s} dv dt \\ &= \int_0^\infty \frac{v^{-s}}{1+v} dv \\ &= \frac{\pi}{\sin \pi(1-s)} \\ &= \frac{\pi}{\sin \pi s}, \end{aligned}$$

and the theorem is proved.

In particular, by putting $s = 1/2$, and noting that $\Gamma(s) > 0$ whenever $s > 0$, we find that

$$\Gamma(1/2) = \sqrt{\pi}.$$

We continue our study of the gamma function by considering its reciprocal, which turns out to be an entire function with remarkably simple properties.

Theorem 1.6 *The function Γ has the following properties:*

- (i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s=0, -1, -2, \dots$ and it vanishes nowhere else.
- (ii) $1/\Gamma(s)$ has growth

$$\left| \frac{1}{\Gamma(s)} \right| \leq c_1 e^{c_2 |s| \log |s|}.$$

Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that

$$\left| \frac{1}{\Gamma(s)} \right| \leq c(\epsilon) e^{c_2 |s|^{1+\epsilon}}.$$

Proof. By the theorem we may write

$$(5) \quad \frac{1}{\Gamma(s)} = \Gamma(1-s) \frac{\sin \pi s}{\pi},$$

so the simple poles of $\Gamma(1-s)$, which are at $s = 1, 2, 3, \dots$ are cancelled by the simple zeros of $\sin \pi s$, and therefore $1/\Gamma$ is entire with simple zeros at $s = 0, -1, -2, -3, \dots$

To prove the estimate, we begin by showing that

$$\int_1^\infty e^{-t} t^\sigma dt \leq e^{(\sigma+1) \log(\sigma+1)}$$

whenever $\sigma = \operatorname{Re}(s)$ is positive. Choose n so that $\sigma \leq n \leq \sigma + 1$. Then

$$\begin{aligned} \int_1^\infty e^{-t} t^\sigma dt &\leq \int_0^\infty e^{-t} t^n dt \\ &= n! \\ &\leq n^n \\ &= e^{n \log n} \\ &\leq e^{(\sigma+1) \log(\sigma+1)}. \end{aligned}$$

Since the relation (3) holds on all of $\mathbb{C}$, we see from (5) that

$$\frac{1}{\Gamma(s)} = \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \right) \frac{\sin \pi s}{\pi} + \left(\int_1^{\infty} e^{-t} t^{-s} dt \right) \frac{\sin \pi s}{\pi}.$$

However, from our previous observation,

$$\left| \int_1^{\infty} e^{-t} t^{-s} dt \right| \leq \int_1^{\infty} e^{-t} t^{|\sigma|} dt \leq e^{(|\sigma|+1) \log(|\sigma|+1)},$$

and because $|\sin \pi s| \leq e^{\pi|s|}$ (by Euler's formula for the sine function) we find that the second term in the formula for $1/\Gamma(s)$ is dominated by $ce^{(|s|+1) \log(|s|+1)} e^{\pi|s|}$, which is itself majorized by $c_1 e^{c_2|s| \log|s|}$. Next, we consider the term

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi}.$$

There are two cases: $|\operatorname{Im}(s)| > 1$ and $|\operatorname{Im}(s)| \leq 1$. In the first case, this expression is dominated in absolute value by $ce^{\pi|s|}$. If $|\operatorname{Im}(s)| \leq 1$, we choose k to be the integer so that $k - 1/2 \leq \operatorname{Re}(s) < k + 1/2$. Then if $k \geq 1$,

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi} &= (-1)^{k-1} \frac{\sin \pi s}{(k-1)!(k-s)\pi} + \\ &+ \sum_{n \neq k-1} (-1)^n \frac{\sin \pi s}{n!(n+1-s)\pi}. \end{aligned}$$

Both terms on the right are bounded; the first because $\sin \pi s$ vanishes at $s = k$, and the second because the sum is majorized by $c \sum 1/n!$.

When $k \leq 0$, then $\operatorname{Re}(s) < 1/2$ by our supposition, and $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)}$ is again bounded by $c \sum 1/n!$. This concludes the proof of the theorem.

The fact that $1/\Gamma$ satisfies the type of growth conditions discussed in Chapter 5 leads naturally to the product formula for the function $1/\Gamma$, which we treat next.

Theorem 1.7 *For all $s \in \mathbb{C}$,*

$$\frac{1}{\Gamma(s)} = e^{\gamma s} \prod_{n=1}^{\infty} \left(1 + \frac{s}{n} \right) e^{-s/n}.$$

The real number γ , which is known as **Euler's constant**, is defined by

$$\gamma = \lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{1}{n} - \log N.$$

The existence of the limit was already proved in Proposition 3.10, Chapter 8 of Book I, but we shall repeat the argument here for completeness. Observe that

$$\sum_{n=1}^N \frac{1}{n} - \log N = \sum_{n=1}^N \frac{1}{n} - \int_1^N \frac{1}{x} dx = \sum_{n=1}^{N-1} \int_n^{n+1} \left[\frac{1}{n} - \frac{1}{x} \right] dx + \frac{1}{N},$$

and by the mean value theorem applied to $f(x) = 1/x$ we have

$$\left| \frac{1}{n} - \frac{1}{x} \right| \leq \frac{1}{n^2} \quad \text{for all } n \leq x \leq n+1.$$

Hence

$$\sum_{n=1}^{\infty} \frac{1}{n} - \log N = \sum_{n=1}^{N-1} a_n + \frac{1}{N}$$

where $|a_n| \leq 1/n^2$. Therefore $\sum a_n$ converges, which proves that the limit defining γ exists. We may now proceed with the proof of the factorization of $1/\Gamma$.

Proof. By the Hadamard factorization theorem and the fact that $1/\Gamma$ is entire, of growth order 1, and has simple zeros at $s = 0, -1, -2, \dots$, we can expand $1/\Gamma$ in a Weierstrass product of the form

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n=1}^{\infty} \left(1 + \frac{s}{n} \right) e^{-s/n}.$$

Here A and B are two constants that are to be determined. Remembering that $s\Gamma(s) \rightarrow 1$ as $s \rightarrow 0$, we find that $B = 0$ (or some integer multiple of $2\pi i$, which of course gives the same result). Putting $s = 1$, and using

the fact that $\Gamma(1) = 1$ yields

$$\begin{aligned}
 e^{-A} &= \prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} e^{\sum_{n=1}^N [\log(1+1/n) - 1/n]} \\
 &= \lim_{N \rightarrow \infty} e^{-(\sum_{n=1}^N 1/n) + \log N + \log(1+1/N)} \\
 &= e^{-\gamma}.
 \end{aligned}$$

Therefore $A = \gamma + 2\pi i k$ for some integer k . Since $\Gamma(s)$ is real whenever s is real, we must have $k = 0$, and the argument is complete.

Note that the proof shows that the function $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function that has:

- (i) simple zeros at $s = 0, -1, -2, \dots$ and vanishes nowhere else, and
- (ii) order of growth ≤ 1 .

Observe that $\sin \pi s$ has a similar characterization (except the zeros are now at *all* the integers). However, while $\sin \pi s$ has a stricter growth estimate of the form $\sin \pi s = O(e^{c|s|})$, this estimate (without the logarithm in the exponent) does not hold for $1/\Gamma(s)$ as Exercise 12 demonstrates.

2 The zeta function

The Riemann **zeta function** is initially defined for real $s > 1$ by the convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

As in the case of the gamma function, ζ can be continued into the complex plane. There are several proofs of this fact, and we present in the next section the one that relies on the functional equation of ζ .

2.1 Functional equation and analytic continuation

In parallel to the gamma function, we first provide a simple extension of ζ to a half-plane in $\mathbb{C}$.

Proposition 2.1 *The series defining $\zeta(s)$ converges for $\operatorname{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.*

Proof. If $s = \sigma + it$ where σ and t are real, then

$$|n^{-s}| = |e^{-s \log n}| = e^{-\sigma \log n} = n^{-\sigma}.$$

As a consequence, if $\sigma > 1 + \delta > 1$ the series defining ζ is uniformly bounded by $\sum_{n=1}^{\infty} 1/n^{1+\delta}$, which converges. Therefore, the series $\sum 1/n^s$ converges uniformly on every half-plane $\operatorname{Re}(s) > 1 + \delta > 1$, and therefore defines a holomorphic function in $\operatorname{Re}(s) > 1$.

The analytic continuation of ζ to a meromorphic function in $\mathbb{C}$ is more subtle than in the case of the gamma function. The proof we present here relates ζ to Γ and another important function.

Consider the **theta function**, already introduced in Chapter 4, which is defined for real $t > 0$ by

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}.$$

An application of the Poisson summation formula (Theorem 2.4 in Chapter 4) gave the functional equation satisfied by ϑ , namely

$$\vartheta(t) = t^{-1/2} \vartheta(1/t).$$

The growth and decay of ϑ we shall need are

$$\vartheta(t) \leq C t^{-1/2} \quad \text{as } t \rightarrow 0,$$

and

$$|\vartheta(t) - 1| \leq C e^{-\pi t} \quad \text{for some } C > 0, \text{ and all } t \geq 1.$$

The inequality for t tending to zero follows from the functional equation, while the behavior as t tends to infinity follows from the fact that

$$\sum_{n \geq 1} e^{-\pi n^2 t} \leq \sum_{n \geq 1} e^{-\pi n t} \leq C e^{-\pi t}$$

for $t \geq 1$.

We are now in a position to prove an important relation among ζ , Γ and ϑ .

Theorem 2.2 *If $\operatorname{Re}(s) > 1$, then*

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2} \int_0^\infty u^{(s/2)-1} [\vartheta(u) - 1] du.$$

Proof. This and further arguments are based on the observation that

$$(6) \quad \int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1} du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \quad \text{if } n \geq 1.$$

Indeed, if we make the change of variables $u = t/\pi n^2$ in the integral, the left-hand side becomes

$$\left(\int_0^\infty e^{-t} t^{(s/2)-1} dt \right) (\pi n^2)^{-s/2},$$

which is precisely $\pi^{-s/2}\Gamma(s/2)n^{-s}$. Next, note that

$$\frac{\vartheta(u) - 1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}.$$

The estimates for ϑ given before the statement of the theorem justify an interchange of the infinite sum with the integral, and thus

$$\begin{aligned} \frac{1}{2} \int_0^\infty u^{(s/2)-1} [\vartheta(u) - 1] du &= \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1} e^{-\pi n^2 u} du \\ &= \pi^{-s/2}\Gamma(s/2) \sum_{n=1}^\infty n^{-s} \\ &= \pi^{-s/2}\Gamma(s/2)\zeta(s), \end{aligned}$$

as was to be shown.

In view of this, we now consider the modification of the ζ function called the **xi function**, which makes the former appear more symmetric. It is defined for $\operatorname{Re}(s) > 1$ by

$$(7) \quad \xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Theorem 2.3 *The function ξ is holomorphic for $\operatorname{Re}(s) > 1$ and has an analytic continuation to all of $\mathbb{C}$ as a meromorphic function with simple poles at $s = 0$ and $s = 1$. Moreover,*

$$\xi(s) = \xi(1-s) \quad \text{for all } s \in \mathbb{C}.$$

Proof. The idea of the proof is to use the functional equation for ϑ , namely

$$\sum_{n=-\infty}^{\infty} e^{-\pi n^2 u} = u^{-1/2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 / u}, \quad u > 0.$$

We then could multiply both sides by $u^{(s/2)-1}$ and try to integrate in u . Disregarding the terms corresponding to $n = 0$ (which produce infinities in both sums), we would get the desired equality once we invoked formula (6), and the parallel formula obtained by making the change of variables $u \mapsto 1/u$. The actual proof requires a little more work and goes as follows.

Let $\psi(u) = [\vartheta(u) - 1]/2$. The functional equation for the theta function, namely $\vartheta(u) = u^{-1/2}\vartheta(1/u)$, implies

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

Now, by Theorem 2.2 for $\operatorname{Re}(s) > 1$, we have

$$\begin{aligned} \pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^\infty u^{(s/2)-1}\psi(u) du \\ &= \int_0^1 u^{(s/2)-1}\psi(u) du + \int_1^\infty u^{(s/2)-1}\psi(u) du \\ &= \int_0^1 u^{(s/2)-1} \left[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2} \right] du + \\ &\quad + \int_1^\infty u^{(s/2)-1}\psi(u) du \\ &= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du \end{aligned}$$

whenever $\operatorname{Re}(s) > 1$. Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du.$$

Since the function ψ has exponential decay at infinity, the integral above defines an entire function, and we conclude that ξ has an analytic continuation to all of $\mathbb{C}$ with simple poles at $s = 0$ and $s = 1$. Moreover, it is

immediate that the integral remains unchanged if we replace s by $1 - s$, and the same is true for the sum of the two terms $1/(s - 1) - 1/s$. We conclude that $\xi(s) = \xi(1 - s)$ as was to be shown.

From the identity we have proved for ξ we obtain the desired result for the zeta function: its analytic continuation and its functional equation.

Theorem 2.4 *The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.*

Proof. A look at (7) provides the meromorphic continuation of ζ , namely

$$\zeta(s) = \pi^{s/2} \frac{\xi(s)}{\Gamma(s/2)}.$$

Recall that $1/\Gamma(s/2)$ is entire with simple zeros at $0, -2, -4, \dots$, so the simple pole of $\xi(s)$ at the origin is cancelled by the corresponding zero of $1/\Gamma(s/2)$. As a consequence, the only singularity of ζ is a simple pole at $s = 1$.

We shall now present a more elementary approach to the analytic continuation of the zeta function, which easily leads to its extension in the half-plane $\operatorname{Re}(s) > 0$. This method will be useful in studying the growth properties of ζ near the line $\operatorname{Re}(s) = 1$ (which will be needed in the next chapter). The idea behind it is to compare the sum $\sum_{n=1}^{\infty} n^{-s}$ with the integral $\int_1^{\infty} x^{-s} dx$.

Proposition 2.5 *There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^{\infty}$ that satisfy the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$, where $s = \sigma + it$, and such that*

$$(8) \quad \sum_{1 \leq n < N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1 \leq n < N} \delta_n(s),$$

whenever N is an integer > 1 .

This proposition has the following consequence.

Corollary 2.6 *For $\operatorname{Re}(s) > 0$ we have*

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^{\infty} \delta_n(s)$ is holomorphic in the half-plane $\operatorname{Re}(s) > 0$.

To prove the proposition we compare $\sum_{1 \leq n < N} n^{-s}$ with $\sum_{1 \leq n < N} \int_n^{n+1} x^{-s} dx$, and set

$$(9) \quad \delta_n(s) = \int_n^{n+1} \left[\frac{1}{n^s} - \frac{1}{x^s} \right] dx.$$

The mean-value theorem applied to $f(x) = x^{-s}$ yields

$$\left| \frac{1}{n^s} - \frac{1}{x^s} \right| \leq \frac{|s|}{n^{\sigma+1}}, \quad \text{whenever } n \leq x \leq n+1.$$

Therefore $|\delta_n(s)| \leq |s|/n^{\sigma+1}$, and since

$$\int_1^N \frac{dx}{x^s} = \sum_{1 \leq n < N} \int_n^{n+1} \frac{dx}{x^s},$$

the proposition is proved.

Turning to the corollary, we assume first that $\operatorname{Re}(s) > 1$. We let N tend to infinity in formula (8) of the proposition, and observe that by the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$ we have the uniform convergence of the series $\sum \delta_n(s)$ (in any half-plane $\operatorname{Re}(s) \geq \delta$ when $\delta > 0$). Since $\operatorname{Re}(s) > 1$, the series $\sum n^{-s}$ converges to $\zeta(s)$, and this proves the assertion when $\operatorname{Re}(s) > 1$. The uniform convergence also shows that $\sum \delta_n(s)$ is holomorphic when $\operatorname{Re}(s) > 0$, and thus shows that $\zeta(s)$ is extendable to that half-plane, and that the identity continues to hold there.

Remark. The idea described above can be developed step by step to yield the continuation of ζ into the entire complex plane, as shown in Problems 2 and 3. Another argument giving the full analytic continuation of ζ is outlined in Exercises 15 and 16.

As an application of the proposition we can show that the growth of $\zeta(s)$ near the line $\operatorname{Re}(s) = 1$ is “mild.” Recall that when $\operatorname{Re}(s) > 1$, we have $|\zeta(s)| \leq \sum_{n=1}^{\infty} n^{-\sigma}$, and so $\zeta(s)$ is bounded in any half-plane $\operatorname{Re}(s) \geq 1 + \delta$, with $\delta > 0$. We shall see that on the line $\operatorname{Re}(s) = 1$, $|\zeta(s)|$ is majorized by $|t|^\epsilon$, for every $\epsilon > 0$, and that the growth near the line is not much worse. The estimates below are not optimal. In fact, they are rather crude but suffice for what is needed later on.

Proposition 2.7 *Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0 , $0 \leq \sigma_0 \leq 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that*

$$(i) \quad |\zeta(s)| \leq c_\epsilon |t|^{1-\sigma_0+\epsilon}, \text{ if } \sigma_0 \leq \sigma \text{ and } |t| \geq 1.$$

(ii) $|\zeta'(s)| \leq c_\epsilon |t|^\epsilon$, if $1 \leq \sigma$ and $|t| \geq 1$.

In particular, the proposition implies that $\zeta(1+it) = O(|t|^\epsilon)$ as $|t|$ tends to infinity,³ and the same estimate also holds for ζ' . For the proof, we use Corollary 2.6. Recall the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$. We also have the estimate $|\delta_n(s)| \leq 2/n^\sigma$, which follows from the expression for $\delta_n(s)$ given by (9) and the fact that $|n^{-s}| = n^{-\sigma}$ and $|x^{-s}| \leq n^{-\sigma}$ if $x \geq n$. We then combine these two estimates for $|\delta_n(s)|$ via the observation that $A = A^\delta A^{1-\delta}$, to obtain the bound

$$|\delta_n(s)| \leq \left(\frac{|s|}{n^{\sigma_0+1}} \right)^\delta \left(\frac{2}{n^{\sigma_0}} \right)^{1-\delta} \leq \frac{2|s|^\delta}{n^{\sigma_0+\delta}},$$

as long as $\delta \geq 0$. Now choose $\delta = 1 - \sigma_0 + \epsilon$ and apply the identity in Corollary 2.6. Then, with $\sigma = \operatorname{Re}(s) \geq \sigma_0$, we find

$$|\zeta(s)| \leq \left| \frac{1}{s-1} \right| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

and conclusion (i) is proved. The second conclusion is actually a consequence of the first by a slight modification of Exercise 8 in Chapter 2. For completeness we sketch the argument. By the Cauchy integral formula,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{i\theta} d\theta,$$

where the integration is taken over a circle of radius r centered at the point s . Now choose $r = \epsilon$ and observe that this circle lies in the half-plane $\operatorname{Re}(s) \geq 1 - \epsilon$, and so (ii) follows as a consequence of (i) on replacing 2ϵ by ϵ .

3 Exercises

1. Prove that

$$\Gamma(s) = \lim_{n \rightarrow \infty} \frac{n^s n!}{s(s+1) \cdots (s+n)}$$

whenever $s \neq 0, -1, -2, \dots$

[Hint: Use the product formula for $1/\Gamma$, and the definition of the Euler constant γ .]

2. Prove that

$$\prod_{n=1}^{\infty} \frac{n(n+a+b)}{(n+a)(n+b)} = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+1)}$$

³The reader should recall the O notation which was introduced at the end of Chapter 1.

whenever a and b are positive. Using the product formula for $\sin \pi s$, give another proof that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$.

3. Show that Wallis's product formula can be written as

$$\sqrt{\frac{\pi}{2}} = \lim_{n \rightarrow \infty} \frac{2^{2n} (n!)^2}{(2n+1)!} (2n+1)^{1/2}.$$

As a result, prove the following identity:

$$\Gamma(s)\Gamma(s+1/2) = \sqrt{\pi} 2^{1-2s} \Gamma(2s).$$

4. Prove that if we take

$$f(z) = \frac{1}{(1-z)^\alpha}, \quad \text{for } |z| < 1$$

(defined in terms of the principal branch of the logarithm), where α is a fixed complex number, then

$$f(z) = \sum_{n=0}^{\infty} a_n(\alpha) z^n$$

with

$$a_n(\alpha) \sim \frac{1}{\Gamma(\alpha)} n^{\alpha-1} \quad \text{as } n \rightarrow \infty.$$

5. Use the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ to prove that

$$|\Gamma(1/2 + it)| = \sqrt{\frac{2\pi}{e^{\pi t} + e^{-\pi t}}}, \quad \text{whenever } t \in \mathbb{R}.$$

6. Show that

$$1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} - \frac{1}{2} \log n \rightarrow \frac{\gamma}{2} + \log 2,$$

where γ is Euler's constant.

7. The **Beta function** is defined for $\operatorname{Re}(\alpha) > 0$ and $\operatorname{Re}(\beta) > 0$ by

$$B(\alpha, \beta) = \int_0^1 (1-t)^{\alpha-1} t^{\beta-1} dt.$$

(a) Prove that $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

(b) Show that $B(\alpha, \beta) = \int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}} du$.

[Hint: For part (a), note that

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty \int_0^\infty t^{\alpha-1} s^{\beta-1} e^{-t-s} dt ds,$$

and make the change of variables $s = ur$, $t = u(1-r)$.]

8. The Bessel functions arise in the study of spherical symmetries and the Fourier transform. See Chapter 6 in Book I. Prove that the following power series identity holds for Bessel functions of real order $\nu > -1/2$:

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1/2)\sqrt{\pi}} \int_{-1}^1 e^{ixt} (1-t^2)^{\nu-(1/2)} dt = \left(\frac{x}{2}\right)^\nu \sum_{m=0}^\infty \frac{(-1)^m \left(\frac{x^2}{4}\right)^m}{m! \Gamma(\nu+m+1)}$$

whenever $x > 0$. In particular, the Bessel function J_ν satisfies the ordinary differential equation

$$\frac{d^2 J_\nu}{dx^2} + \frac{1}{x} \frac{dJ_\nu}{dx} + \left(1 - \frac{\nu^2}{x^2}\right) J_\nu = 0.$$

[Hint: Expand the exponential e^{ixt} in a power series, and express the remaining integrals in terms of the gamma function, using Exercise 7.]

9. The hypergeometric series $F(\alpha, \beta, \gamma; z)$ was defined in Exercise 16 of Chapter 1. Show that

$$F(\alpha, \beta, \gamma; z) = \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \int_0^1 t^{\beta-1} (1-t)^{\gamma-\beta-1} (1-zt)^{-\alpha} dt.$$

Here $\alpha > 0, \beta > 0, \gamma > \beta$, and $|z| < 1$.

Show as a result that the hypergeometric function, initially defined by a power series convergent in the unit disc, can be continued analytically to the complex plane slit along the half-line $[1, \infty)$.

Note that

$$\log(1-z) = -zF(1, 1, 2; z),$$

$$e^z = \lim_{\beta \rightarrow \infty} F(1, \beta, 1; z/\beta),$$

$$(1-z)^{-\alpha} = F(\alpha, 1, 1; z).$$

[Hint: To prove the integral identity, expand $(1-zt)^{-\alpha}$ as a power series.]

10. An integral of the form

$$F(z) = \int_0^\infty f(t) t^{z-1} dt$$

is called a **Mellin transform**, and we shall write $\mathcal{M}(f)(z) = F(z)$. For example, the gamma function is the Mellin transform of the function e^{-t} .

(a) Prove that

$$\mathcal{M}(\cos)(z) = \int_0^\infty \cos(t)t^{z-1} dt = \Gamma(z) \cos\left(\pi \frac{z}{2}\right) \quad \text{for } 0 < \operatorname{Re}(z) < 1,$$

and

$$\mathcal{M}(\sin)(z) = \int_0^\infty \sin(t)t^{z-1} dt = \Gamma(z) \sin\left(\pi \frac{z}{2}\right) \quad \text{for } 0 < \operatorname{Re}(z) < 1.$$

(b) Show that the second of the above identities is valid in the larger strip $-1 < \operatorname{Re}(z) < 1$, and that as a consequence, one has

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2} \quad \text{and} \quad \int_0^\infty \frac{\sin x}{x^{3/2}} dx = \sqrt{2\pi}.$$

This generalizes the calculation in Exercise 2 of Chapter 2.

[Hint: For the first part, consider the integral of the function $f(w) = e^{-w}w^{z-1}$ around the contour illustrated in Figure 1. Use analytic continuation to prove the second part.]

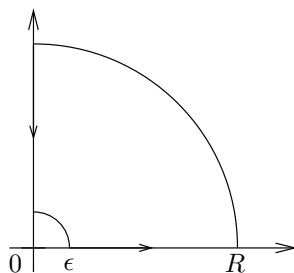


Figure 1. The contour in Exercise 10

11. Let $f(z) = e^{az}e^{-e^z}$ where $a > 0$. Observe that in the strip $\{x + iy : |y| < \pi\}$ the function $f(x + iy)$ is exponentially decreasing as $|x|$ tends to infinity. Prove that

$$\hat{f}(\xi) = \Gamma(a + i\xi), \quad \text{for all } \xi \in \mathbb{R}.$$

12. This exercise gives two simple observations about $1/\Gamma$.

- (a) Show that $1/|\Gamma(s)|$ is not $O(e^{c|s|})$ for any $c > 0$. [Hint: If $s = -k - 1/2$, where k is a positive integer, then $|1/\Gamma(s)| \geq k!/\pi$.]
- (b) Show that there is no entire function $F(s)$ with $F(s) = O(e^{c|s|})$ that has simple zeros at $s = 0, -1, -2, \dots, -n, \dots$, and that vanishes nowhere else.

13. Prove that

$$\frac{d^2 \log \Gamma(s)}{ds^2} = \sum_{n=0}^{\infty} \frac{1}{(s+n)^2}$$

whenever s is a positive number. Show that if the left-hand side is interpreted as $(\Gamma'/\Gamma)'$, then the above formula also holds for all complex numbers s with $s \neq 0, -1, -2, \dots$

14. This exercise gives an asymptotic formula for $\log n!$. A more refined asymptotic formula for $\Gamma(s)$ as $s \rightarrow \infty$ (Stirling's formula) is given in Appendix A.

(a) Show that

$$\frac{d}{dx} \int_x^{x+1} \log \Gamma(t) dt = \log x, \quad \text{for } x > 0,$$

and as a result

$$\int_x^{x+1} \log \Gamma(t) dt = x \log x - x + c.$$

- (b) Show as a consequence that $\log \Gamma(n) \sim n \log n$ as $n \rightarrow \infty$. In fact, prove that $\log \Gamma(n) \sim n \log n + O(n)$ as $n \rightarrow \infty$. [Hint: Use the fact that $\Gamma(x)$ is monotonically increasing for all large x .]

15. Prove that for $\operatorname{Re}(s) > 1$,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

[Hint: Write $1/(e^x - 1) = \sum_{n=1}^{\infty} e^{-nx}$.]

16. Use the previous exercise to give another proof that $\zeta(s)$ is continuable in the complex plane with only singularity a simple pole at $s = 1$.

[Hint: Write

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^1 \frac{x^{s-1}}{e^x - 1} dx + \frac{1}{\Gamma(s)} \int_1^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

The second integral defines an entire function, while

$$\int_0^1 \frac{x^{s-1}}{e^x - 1} dx = \sum_{m=0}^{\infty} \frac{B_m}{m!(s+m-1)},$$

where B_m denotes the m^{th} Bernoulli number defined by

$$\frac{x}{e^x - 1} = \sum_{m=0}^{\infty} \frac{B_m}{m!} x^m.$$

Then $B_0 = 1$, and since $z/(e^z - 1)$ is holomorphic for $|z| < 2\pi$, we must have $\limsup_{m \rightarrow \infty} |B_m/m!|^{1/m} = 1/2\pi$.]

17. Let f be an indefinitely differentiable function on $\mathbb{R}$ that has compact support, or more generally, let f belong to the Schwartz space.⁴ Consider

$$I(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} f(x) x^{-1+s} dx.$$

- (a) Observe that $I(s)$ is holomorphic for $\text{Re}(s) > 0$. Prove that I has an analytic continuation as an entire function in the complex plane.
- (b) Prove that $I(0) = 0$, and more generally

$$I(-n) = (-1)^n f^{(n+1)}(0) \quad \text{for all } n \geq 0.$$

[Hint: To prove the analytic continuation, as well as the formulas in the second part, integrate by parts to show that $I(s) = \frac{(-1)^k}{\Gamma(s+k)} \int_0^{\infty} f^{(k)}(x) x^{s+k-1} dx$.]

4 Problems

1. This problem provides further estimates for ζ and ζ' near $\text{Re}(s) = 1$.

- (a) Use Proposition 2.5 and its corollary to prove

$$\zeta(s) = \sum_{1 \leq n < N} n^{-s} - \frac{N^{s-1}}{s-1} + \sum_{n \geq N} \delta_n(s)$$

for every integer $N \geq 2$, whenever $\text{Re}(s) > 0$.

- (b) Show that $|\zeta(1+it)| = O(\log |t|)$, as $|t| \rightarrow \infty$ by using the previous result with $N = \text{greatest integer in } |t|$.

⁴The Schwartz space on $\mathbb{R}$ is denoted by $\mathcal{S}$ and consists of all indefinitely differentiable functions f , so that f and all its derivatives decay faster than any polynomials. In other words, $\sup_{x \in \mathbb{R}} |x|^m |f^{(\ell)}(x)| < \infty$ for all integers $m, \ell \geq 0$. This space appeared in the study of the Fourier transform in Book I.

- (c) The second conclusion of Proposition 2.7 can be similarly refined.
- (d) Show that if $t \neq 0$ and t is fixed, then the partial sums of the series $\sum_{n=1}^{\infty} 1/n^{1+it}$ are bounded, but the series does not converge.

2.* Prove that for $\operatorname{Re}(s) > 0$

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx$$

where $\{x\}$ is the fractional part of x .

3.* If $Q(x) = \{x\} - 1/2$, then we can write the expression in the previous problem as

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s \int_1^{\infty} \frac{Q(x)}{x^{s+1}} dx.$$

Let us construct $Q_k(x)$ recursively so that

$$\int_0^1 Q_k(x) dx = 0, \quad \frac{dQ_{k+1}}{dx} = Q_k(x), \quad Q_0(x) = Q(x) \quad \text{and} \quad Q_k(x+1) = Q_k(x).$$

Then we can write

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s \int_1^{\infty} \left(\frac{d^k}{dx^k} Q_k(x) \right) x^{-s-1} dx,$$

and a k -fold integration by parts gives the analytic continuation for $\zeta(s)$ when $\operatorname{Re}(s) > -k$.

4.* The functions Q_k in the previous problem are related to the Bernoulli polynomials $B_k(x)$ by the formula

$$Q_k(x) = \frac{B_{k+1}(x)}{(k+1)!} \quad \text{for } 0 \leq x \leq 1.$$

Also, if k is a positive integer, then

$$2\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k}}{(2k)!} B_{2k},$$

where $B_k = B_k(0)$ are the Bernoulli numbers. For the definition of $B_k(x)$ and B_k see Chapter 3 in Book I.

7 The Zeta Function and Prime Number Theorem

Bernhard Riemann, whose extraordinary intuitive powers we have already mentioned, has especially renovated our knowledge of the distribution of prime numbers, also one of the most mysterious questions in mathematics. He has taught us to deduce results in that line from considerations borrowed from the integral calculus: more precisely, from the study of a certain quantity, a function of a variable s which may assume not only real, but also imaginary values. He proved some important properties of that function, but enunciated two or three as important ones without giving the proof. At the death of Riemann, a note was found among his papers, saying “These properties of $\zeta(s)$ (the function in question) are deduced from an expression of it which, however, I did not succeed in simplifying enough to publish it.”

We still have not the slightest idea of what the expression could be. As to the properties he simply enunciated, some thirty years elapsed before I was able to prove all of them but one. The question concerning that last one remains unsolved as yet, though, by an immense labor pursued throughout this last half century, some highly interesting discoveries in that direction have been achieved. It seems more and more probable, but still not at all certain, that the “Riemann hypothesis” is true.

J. Hadamard, 1945

Euler found, through his product formula for the zeta function, a deep connection between analytical methods and arithmetic properties of numbers, in particular primes. An easy consequence of Euler’s formula is that the sum of the reciprocals of all primes, $\sum_p 1/p$, diverges, a result that quantifies the fact that there are infinitely many prime numbers. The natural problem then becomes that of understanding how these primes are distributed. With this in mind, we consider the